

3(a)	(cell type A) phloem sieve tube element ; <i>plus any two from:</i> connect the cytoplasm of the two cells / <i>ref. to</i> the symplast pathway ; <i>idea of</i> allow (rapid) diffusion of, sucrose / named molecules, from one cell to an adjacent cell / AW ; <i>idea of</i> efficiency of, loading into / unloading from, sieve tubes ;	3
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Question	Answer	Marks															
3(b)	<table> <tr> <th>feature of polysaccharide</th><th>callose molecule</th><th>cellulose molecule</th></tr> <tr> <td>monosaccharide used to synthesise the polysaccharide</td><td></td><td>beta / β, glucose ; R B glucose</td></tr> <tr> <td>bond connecting the monomers</td><td>1, 3 glycosidic bond ;</td><td></td></tr> <tr> <td>shape of molecule</td><td></td><td>straight / linear ;</td></tr> <tr> <td>orientation of monosaccharides in the molecule</td><td>adjacent glucose molecules same way up / AW ;</td><td></td></tr> </table>	feature of polysaccharide	callose molecule	cellulose molecule	monosaccharide used to synthesise the polysaccharide		beta / β , glucose ; R B glucose	bond connecting the monomers	1, 3 glycosidic bond ;		shape of molecule		straight / linear ;	orientation of monosaccharides in the molecule	adjacent glucose molecules same way up / AW ;		4
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3(c)	any three from: (hydrogen bonding) gives adhesion between water molecules and, cellulose / hydrophilic parts of lignin (in xylem wall) ; (hydrogen bonding) gives cohesion between water molecules ; formation of continuous column / AW ; A adhesion, prevents column of water collapsing / AW transpiration pull ; A water, drawn up / pulled up by transpiration / pulled up by evaporation detail ; e.g. water evaporating from spongy mesophyll surfaces water vapour lost from, leaf / plant, during transpiration <i>idea of</i> creates tension AVP ; <i>if mp 1 and mp 2 are not awarded, allow one mark for reference to cohesion <u>and</u> adhesion</i>	3															

Question	Answer	Marks
3(d)	any two from: hydrogen bonds between water molecules must break for water, to change state / evaporate / AW ; loss of, heat / energy for, evaporation / transpiration (will cool the, leaf / plant) ;	2